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Aproximación en norma $\mathcal{L}^p$ de la indicatriz por funciones continuas con soporte compacto

Lema 1. *Sea $A \in \mathcal{B}(\mathbb{R})$ tal que $m(A) < \infty$ y sea $p \in [1, \infty)$*

$$\implies \forall \varepsilon > 0 : \exists g \in \mathcal{C}_c(\mathbb{R}) : \|\mathbb{1}_A - g\|_p < \varepsilon.$$

***Demostración*:** Como m es regular, existe K compacto y V abierto tales que $K \subset A \subset V$ y $m(V \setminus K) < \varepsilon^p$. Por el `Lem-aprox-indicatriz-compacto-abierto-continua`/Lema 1, $\exists g \in \mathcal{C}_c(\mathbb{R}) : \forall x \in \mathbb{R} : \mathbb{1}_K(x) \leq g(x) \leq \mathbb{1}_V(x)$. Por tanto, tenemos que

$$\|\mathbb{1}_A - g\|_p^p = \int_{\mathbb{R}} |\mathbb{1}_A(x) - g(x)|^p \,\mathrm{d}x \leq \int_{\mathbb{R}} |\mathbb{1}_V(x) - \mathbb{1}_K(x)|^p \,\mathrm{d}x = m(V \setminus K) < \varepsilon^p.$$

Luego concluimos que $\|\mathbb{1}_A - g\|_p < \varepsilon$. ∎

Referenciado en

- `Teo-fn-continua-soporte-compacto-denso-lp`
- `Teo-fn-suave-soporte-compacto-denso-lp`

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